

Development & Integration

Date	2 October 2026
Team ID	SWTID-2026-6582
Project Name	PocketSmart AI: Your Smart Budget & Recommendation Assistant
Maximum Marks	Project Deliverable

Frontend Development

The frontend uses Jinja2 templates with HTML, CSS and JavaScript. It includes landing, authentication, dashboard, planner forms, result cards and history views.

Backend Logic Implementation

FastAPI routers handle HTTP requests. Services separate Gemini integration, budget calculations, image handling, product search-link generation and recommendation orchestration.

Gemini API Integration

`app/services/gemini_service.py` initializes the official Google GenAI client using environment configuration and calls Gemini 2.5 Flash Lite for Home, Party and Jewelry generation, including multimodal Jewelry requests.

Database Integration

SQLAlchemy ORM connects to SQLite. The users table stores account metadata and bcrypt password hashes. The recommendations table stores authenticated users' generated plans and budget metadata.

Modular Code Structure

Key modules are organized under `app/routers`, `app/services`, `app/models` and `app/utils`, with templates, static assets and tests separated from backend code.

Error Handling

Invalid form values, malformed AI output, image validation failures and Gemini failures return controlled errors. A failed Gemini call is not replaced with a static plan labelled as AI-generated.

API Key Security

GEMINI_API_KEY and other secrets are loaded from .env/environment settings and are excluded from source control through .gitignore. Secrets are not displayed in the UI.